DeterminingTheNumberOfFactors

Alissa McGill

2026-11-09

The data set for this assignment includes over 19,000 participants data on the Big 5 Personality inventory. The data are from several countries, and many different sources. Your assignment is to determine how many personality factors are represented by this inventory and provide me with all reasons, explanations, and data you used to determine the number of factors that underly the construct "personality." This should be conducted in an exploratory fashion.

```
library(haven)
```

```
## Warning: package 'haven' was built under R version 4.6.1
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(psych)
library(nFactors)
```

```
## Warning: package 'nFactors' was built under R version 4.6.1
```

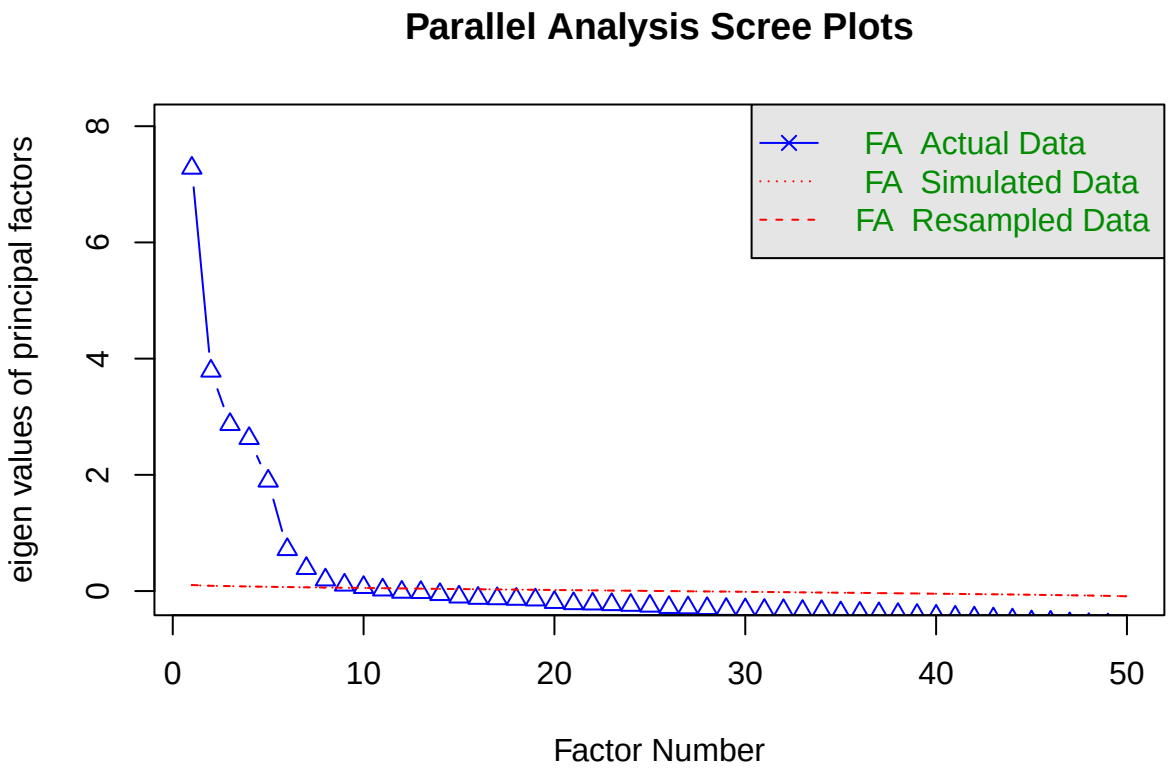
```
Big5Data = read_sav("big5data.sav")
Big5Data  =  select(Big5Data, -c(race, age, engnat, gender, hand, source, country))
```

```
#Calculates Bartlett's test
CorBig5<-cor(Big5Data)
cortest.bartlett(CorBig5, n=19719)
```

```
## $chisq
## [1] 376850.5
##
```

```
## $p.value
## [1] 0
##
## $df
## [1] 1225
```

```
#  parallel analysis
parallelAnalysis = fa.parallel(Big5Data, fa = "fa", plot = TRUE)
```




```
## Parallel analysis suggests that the number of factors =  10  and the number of components =  NA
```

```
parallelAnalysis
```

```
## Call: fa.parallel(x = Big5Data, fa = "fa", plot = TRUE)
## Parallel analysis suggests that the number of factors =  10  and the number of components =  NA
##
##  Eigen Values of
##
##  eigen values of factors
##  [1]  7.28  3.79  2.87  2.63  1.89  0.72  0.39  0.19  0.10  0.06  0.02 -0.02
## [13] -0.02 -0.06 -0.10 -0.12 -0.13 -0.14 -0.15 -0.19 -0.22 -0.22 -0.23 -0.24
## [25] -0.26 -0.28 -0.29 -0.30 -0.31 -0.32 -0.33 -0.33 -0.35 -0.35 -0.37 -0.38
## [37] -0.39 -0.40 -0.42 -0.43 -0.45 -0.46 -0.48 -0.49 -0.53 -0.53 -0.56 -0.57
## [49] -0.58 -0.65
```

```
##
##  eigen values of simulated factors
##  [1]  0.10  0.09  0.08  0.08  0.07  0.07  0.06  0.06  0.06  0.05  0.05  0.05
## [13]  0.04  0.04  0.03  0.03  0.03  0.02  0.02  0.02  0.02  0.01  0.01  0.01
## [25]  0.00  0.00  0.00 -0.01 -0.01 -0.01 -0.02 -0.02 -0.02 -0.03 -0.03 -0.03
## [37] -0.04 -0.04 -0.04 -0.05 -0.05 -0.05 -0.06 -0.06 -0.06 -0.07 -0.07 -0.08
## [49] -0.08 -0.09
##
##  eigen values of components
##  [1] 8.05 4.62 3.75 3.55 2.76 1.58 1.33 1.05 0.97 0.93 0.90 0.86 0.85 0.80 0.79
## [16] 0.73 0.73 0.72 0.68 0.66 0.66 0.64 0.61 0.59 0.58 0.57 0.55 0.54 0.52 0.51
## [31] 0.50 0.49 0.49 0.48 0.45 0.45 0.44 0.42 0.42 0.40 0.40 0.38 0.37 0.36 0.35
## [46] 0.34 0.33 0.32 0.31 0.22
##
##  eigen values of simulated components
## [1] NA
```

```
#this parallel analysis tells me its 10 factors, i don't really think that makes a lot of sense

#oblimin becasue they are probably correlated

# #do the factor analysis for 1
# fitBig5_oneFac<-fa(Big5Data,nfactors = 1, rotate = "oblimin", fm = 'pa')
# #displays results
# fitBig5_oneFac
# print(fitBig5_oneFac$loadings, cutoff = 0.3, sort = TRUE)
#
# #do the factor analysis for 2
# fitBig5_twoFac<-fa(Big5Data,nfactors = 2, rotate = "oblimin", fm = 'pa')
# fitBig5_twoFac
# print(fitBig5_twoFac$loadings, cutoff = 0.3, sort = TRUE)
#
# #do the factor analysis for 3
# fitBig5_threeFac<-fa(Big5Data,nfactors = 3, rotate = "oblimin", fm = 'pa')
# fitBig5_threeFac
# print(fitBig5_threeFac$loadings, cutoff = 0.3, sort = TRUE)
#


#do the factor analysis for 10
fitBig5_tenFac<-fa(Big5Data,nfactors = 10, rotate = "oblimin", fm = 'pa')
```

```
## Loading required namespace: GPArotation
```

```
fitBig5_tenFac
```

```
## Factor Analysis using method =  pa
## Call: fa(r = Big5Data, nfactors = 10, rotate = "oblimin", fm = "pa")
## Standardized loadings (pattern matrix) based upon correlation matrix
##      PA10   PA8   PA5   PA7   PA3   PA2   PA9   PA6   PA4   PA1   h2   u2 com
## E1  -0.65  0.03  0.03  0.09 -0.12  0.06  0.00  0.08  0.09  0.04 0.48 0.52 1.2
## E2   0.72  0.04 -0.03  0.06 -0.17 -0.02 -0.01  0.06  0.00  0.09 0.52 0.48 1.2
## E3  -0.58  0.10 -0.02 -0.02 -0.22 -0.03 -0.05  0.16  0.03 -0.13 0.60 0.40 1.7
```

```
## E4   0.77  0.02  0.06  0.08 -0.06  0.05  0.01  0.11  0.07 -0.04 0.57 0.43 1.1
## E5  -0.69  0.05 -0.01  0.01 -0.02  0.02  0.02  0.12  0.03 -0.12 0.60 0.40 1.1
## E6   0.58  0.04  0.03  0.11 -0.15 -0.09 -0.08  0.05  0.12  0.14 0.47 0.53 1.6
## E7  -0.69  0.02  0.01  0.07 -0.12  0.00  0.01  0.10  0.06 -0.11 0.58 0.42 1.2
## E8   0.62  0.02 -0.01 -0.03 -0.03 -0.03  0.03  0.11  0.03 -0.14 0.35 0.65 1.2
## E9  -0.63  0.03  0.05  0.07 -0.07  0.06  0.00  0.05 -0.02  0.16 0.42 0.58 1.2
## E10  0.67  0.02  0.07  0.04  0.04  0.03 -0.02  0.10  0.04  0.00 0.48 0.52 1.1
## N1   0.02  0.04  0.07  0.10  0.65  0.00  0.00  0.04  0.09 -0.02 0.56 0.44 1.1
## N2   0.02  0.04  0.11 -0.08 -0.60  0.07  0.00  0.14  0.06  0.00 0.42 0.58 1.3
## N3   0.10  0.08  0.06  0.10  0.56  0.04 -0.03  0.14  0.06 -0.11 0.48 0.52 1.5
## N4  -0.10  0.05  0.02 -0.21 -0.24  0.02 -0.04  0.12  0.13  0.15 0.20 0.80 4.6
## N5  -0.01  0.09  0.13  0.18  0.34  0.01 -0.05  0.01  0.15  0.14 0.34 0.66 3.0
## N6  -0.02  0.13  0.11  0.28  0.49 -0.01 -0.03  0.03  0.10  0.13 0.58 0.42 2.2
## N7  -0.01 -0.01  0.00  0.80  0.05  0.02 -0.02  0.02 -0.01 -0.02 0.67 0.33 1.0
## N8  -0.01  0.02 -0.04  0.88  0.03 -0.01  0.01 -0.03 -0.01  0.01 0.78 0.22 1.0
## N9  -0.03 -0.11  0.09  0.31  0.44  0.00  0.02  0.11  0.10  0.08 0.53 0.47 2.4
## N10  0.20  0.02  0.05  0.46  0.24 -0.01  0.07 -0.06 -0.06 -0.02 0.49 0.51 2.1
## A1   0.00 -0.28  0.04  0.17 -0.14  0.05 -0.12  0.13  0.10  0.23 0.27 0.73 4.6
## A2  -0.22  0.24  0.09  0.03 -0.09  0.05  0.00  0.13  0.00 -0.41 0.48 0.52 2.8
## A3  -0.12 -0.44  0.22  0.15  0.08  0.05  0.08  0.11  0.10 -0.03 0.32 0.68 2.5
## A4   0.04  0.78  0.01 -0.01  0.01 -0.01  0.02 -0.02  0.00 -0.08 0.65 0.35 1.0
## A5   0.07 -0.42  0.00  0.02 -0.05  0.02  0.01  0.07  0.09  0.36 0.49 0.51 2.2
## A6   0.04  0.70  0.02  0.00  0.08  0.02 -0.05 -0.02  0.07  0.09 0.46 0.54 1.1
## A7   0.21 -0.27  0.00  0.05 -0.01 -0.04  0.04  0.02  0.06  0.50 0.60 0.40 2.1
## A8  -0.02  0.46  0.01  0.00 -0.06  0.00  0.02  0.11  0.01 -0.21 0.38 0.62 1.6
## A9  -0.04  0.74 -0.01  0.07  0.01  0.03  0.01  0.01 -0.02  0.01 0.58 0.42 1.0
## A10 -0.25  0.38 -0.03 -0.01 -0.16  0.06 -0.01  0.15  0.02  0.03 0.36 0.64 2.6
## C1   0.01 -0.03 -0.39 -0.07  0.01 -0.02  0.07  0.39 -0.01 -0.02 0.40 0.60 2.2
## C2  -0.03  0.02  0.68 -0.08  0.02 -0.02  0.05  0.08 -0.03 -0.03 0.42 0.58 1.1
## C3   0.11  0.00 -0.20  0.01  0.03  0.10  0.05  0.38 -0.10 -0.10 0.27 0.73 2.3
## C4   0.06 -0.05  0.57  0.16  0.09  0.02  0.02 -0.03  0.08 -0.06 0.49 0.51 1.3
## C5  -0.06  0.11 -0.53  0.02 -0.04 -0.02 -0.05  0.23  0.06  0.09 0.43 0.57 1.6
## C6   0.01  0.04  0.73 -0.06  0.05 -0.01 -0.02  0.05  0.00  0.05 0.50 0.50 1.0
## C7   0.07 -0.04 -0.37 -0.06  0.18  0.00  0.06  0.31  0.06 -0.08 0.31 0.69 2.8
## C8   0.06 -0.09  0.39  0.22 -0.08  0.02  0.00 -0.11  0.07  0.03 0.32 0.68 2.2
## C9  -0.03  0.03 -0.42 -0.09  0.16 -0.03 -0.01  0.36  0.08 -0.04 0.42 0.58 2.5
## C10  0.02  0.05 -0.20 -0.06  0.08  0.00  0.06  0.44 -0.11  0.04 0.33 0.67 1.8
## O1  -0.01  0.03 -0.03 -0.06 -0.01  0.03  0.81 -0.02  0.01  0.00 0.66 0.34 1.0
## O2   0.01  0.02 -0.02 -0.04  0.04 -0.01 -0.04 -0.04  0.76 -0.03 0.61 0.39 1.0
## O3   0.08  0.01  0.03  0.05  0.01  0.71  0.06 -0.03  0.06 -0.04 0.50 0.50 1.1
## O4  -0.03 -0.01 -0.05 -0.04 -0.02 -0.11  0.03  0.01  0.61  0.08 0.42 0.58 1.1
## O5  -0.14  0.04  0.02 -0.08 -0.01  0.40  0.12  0.27 -0.13  0.19 0.48 0.52 3.2
## O6   0.01  0.01  0.09  0.01  0.02 -0.73  0.04  0.14  0.05  0.05 0.52 0.48 1.1
## O7  -0.02  0.05 -0.01  0.02 -0.13  0.03  0.21  0.31 -0.30  0.11 0.36 0.64 3.5
## O8   0.00 -0.02  0.04  0.08 -0.01 -0.02  0.80 -0.02  0.00  0.01 0.63 0.37 1.0
## O9   0.18  0.14  0.05  0.03  0.14  0.13  0.14  0.14 -0.13 -0.07 0.19 0.81 7.3
## O10 -0.10  0.01  0.07 -0.05  0.01  0.55  0.07  0.19 -0.15  0.10 0.56 0.44 1.7
##
##                        PA10  PA8  PA5  PA7  PA3  PA2  PA9  PA6  PA4  PA1
## SS loadings            4.96 3.03 2.71 2.62 2.53 1.78 1.59 1.53 1.51 1.28
## Proportion Var         0.10 0.06 0.05 0.05 0.05 0.04 0.03 0.03 0.03 0.03
## Cumulative Var         0.10 0.16 0.21 0.27 0.32 0.35 0.38 0.41 0.45 0.47
## Proportion Explained   0.21 0.13 0.12 0.11 0.11 0.08 0.07 0.06 0.06 0.05
## Cumulative Proportion  0.21 0.34 0.45 0.57 0.67 0.75 0.82 0.88 0.95 1.00
```

```
##
##  With factor correlations of
##       PA10   PA8   PA5   PA7   PA3   PA2   PA9   PA6   PA4   PA1
## PA10  1.00 -0.20  0.05  0.16  0.22 -0.17 -0.05 -0.14  0.10  0.23
## PA8  -0.20  1.00 -0.06 -0.04  0.04  0.12 -0.07  0.17 -0.04 -0.44
## PA5   0.05 -0.06  1.00  0.33  0.11  0.08  0.06 -0.25  0.10  0.05
## PA7   0.16 -0.04  0.33  1.00  0.52  0.02 -0.02 -0.04  0.24  0.15
## PA3   0.22  0.04  0.11  0.52  1.00 -0.04  0.02 -0.04  0.16 -0.01
## PA2  -0.17  0.12  0.08  0.02 -0.04  1.00  0.36  0.16 -0.40 -0.01
## PA9  -0.05 -0.07  0.06 -0.02  0.02  0.36  1.00  0.17 -0.43  0.00
## PA6  -0.14  0.17 -0.25 -0.04 -0.04  0.16  0.17  1.00  0.01  0.02
## PA4   0.10 -0.04  0.10  0.24  0.16 -0.40 -0.43  0.01  1.00  0.10
## PA1   0.23 -0.44  0.05  0.15 -0.01 -0.01  0.00  0.02  0.10  1.00
##
## Mean item complexity =  1.9
## Test of the hypothesis that 10 factors are sufficient.
##
## df null model =  1225  with the objective function =  19.13 with Chi Square =  376850.5
## df of  the model are 770  and the objective function was  1.11
##
## The root mean square of the residuals (RMSR) is  0.02
## The df corrected root mean square of the residuals is  0.02
##
## The harmonic n.obs is  19719 with the empirical chi square  6616.18  with prob <  0
## The total n.obs was  19719  with Likelihood Chi Square =  21885  with prob <  0
##
## Tucker Lewis Index of factoring reliability =  0.911
## RMSEA index =  0.037  and the 90 % confidence intervals are  0.037 0.038
## BIC =  14270.21
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                 PA10  PA8  PA5  PA7  PA3  PA2
## Correlation of (regression) scores with factors   0.92 0.84 0.85 0.82 0.79 0.79
## Multiple R square of scores with factors          0.84 0.71 0.72 0.67 0.62 0.63
## Minimum correlation of possible factor scores     0.69 0.41 0.44 0.33 0.25 0.26
##                                                  PA9  PA6  PA4  PA1
## Correlation of (regression) scores with factors   0.82 0.77 0.75 0.71
## Multiple R square of scores with factors          0.67 0.59 0.56 0.51
## Minimum correlation of possible factor scores     0.34 0.17 0.13 0.02
```

```
print(fitBig5_tenFac$loadings, cutoff = 0.3, sort = FALSE)
```

```
##
## Loadings:
##     PA10   PA8    PA5    PA7    PA3    PA2    PA9    PA6    PA4    PA1
## E1  -0.645
## E2   0.719
## E3  -0.577
## E4   0.768
## E5  -0.689
## E6   0.583
## E7  -0.687
## E8   0.620
## E9  -0.628
```

```
## E10  0.675
## N1                              0.646
## N2                             -0.601
## N3                              0.562
## N4
## N5                              0.341
## N6                              0.494
## N7                       0.798
## N8                       0.881
## N9                       0.312  0.439
## N10                      0.458
## A1
## A2                                                                -0.414
## A3         -0.441
## A4          0.781
## A5         -0.420                                                  0.363
## A6          0.702
## A7                                                                  0.502
## A8          0.458
## A9          0.742
## A10         0.384
## C1                -0.387                          0.385
## C2                 0.682
## C3                                                 0.382
## C4                 0.567
## C5                -0.527
## C6                 0.731
## C7                -0.370                          0.307
## C8                 0.388
## C9                -0.415                          0.357
## C10                                               0.440
## O1                                          0.809
## O2                                                      0.761
## O3                                    0.712
## O4                                                      0.611
## O5                                    0.397
## O6                                   -0.726
## O7                                                0.308 -0.301
## O8                                          0.796
## O9
## O10                                   0.554
##
##                 PA10   PA8   PA5   PA7   PA3   PA2   PA9   PA6   PA4   PA1
## SS loadings    4.723 2.720 2.450 2.134 2.094 1.583 1.447 1.293 1.301 0.981
## Proportion Var 0.094 0.054 0.049 0.043 0.042 0.032 0.029 0.026 0.026 0.020
## Cumulative Var 0.094 0.149 0.198 0.241 0.282 0.314 0.343 0.369 0.395 0.414
```

```
#do the factor analysis for 5
fitBig5_fiveFac<-fa(Big5Data,nfactors = 5, rotate = "oblimin", fm = 'pa')
fitBig5_fiveFac
```

```
## Factor Analysis using method =  pa
## Call: fa(r = Big5Data, nfactors = 5, rotate = "oblimin", fm = "pa")
## Standardized loadings (pattern matrix) based upon correlation matrix
```

```
##       PA1   PA3   PA5   PA2   PA4   h2   u2 com
## E1   0.69  0.04 -0.03 -0.01  0.00 0.46 0.54 1.0
## E2  -0.70 -0.08 -0.04  0.04  0.00 0.48 0.52 1.0
## E3   0.63 -0.17  0.16  0.09 -0.06 0.57 0.43 1.3
## E4  -0.72  0.06  0.04  0.00  0.03 0.52 0.48 1.0
## E5   0.72  0.03  0.12  0.07  0.03 0.60 0.40 1.1
## E6  -0.54  0.02 -0.08  0.00 -0.19 0.40 0.60 1.3
## E7   0.74  0.00  0.06  0.02 -0.01 0.57 0.43 1.0
## E8  -0.60 -0.04  0.11  0.07  0.00 0.33 0.67 1.1
## E9   0.64  0.05 -0.09 -0.02  0.09 0.40 0.60 1.1
## E10 -0.65  0.10  0.03  0.00  0.01 0.45 0.55 1.1
## N1  -0.06  0.69  0.10  0.05 -0.05 0.49 0.51 1.1
## N2   0.07 -0.50 -0.01 -0.09  0.05 0.26 0.74 1.1
## N3  -0.12  0.61  0.20  0.10  0.01 0.43 0.57 1.3
## N4   0.13 -0.30 -0.06  0.07 -0.07 0.14 0.86 1.8
## N5   0.01  0.53  0.01 -0.05 -0.11 0.32 0.68 1.1
## N6   0.00  0.75  0.06 -0.01 -0.07 0.57 0.43 1.0
## N7   0.06  0.71 -0.05 -0.08  0.02 0.52 0.48 1.1
## N8   0.05  0.74 -0.05 -0.08  0.01 0.57 0.43 1.0
## N9   0.03  0.73 -0.15  0.04  0.00 0.54 0.46 1.1
## N10 -0.21  0.58  0.02 -0.10  0.08 0.47 0.53 1.4
## A1   0.05  0.09 -0.44  0.02 -0.07 0.20 0.80 1.2
## A2   0.28 -0.04  0.50 -0.05  0.06 0.41 0.59 1.6
## A3   0.17  0.27 -0.41 -0.15  0.10 0.27 0.73 2.6
## A4  -0.06  0.03  0.80  0.00 -0.01 0.62 0.38 1.0
## A5  -0.06  0.04 -0.66  0.05  0.00 0.45 0.55 1.0
## A6  -0.05  0.13  0.61  0.02 -0.08 0.37 0.63 1.1
## A7  -0.23  0.10 -0.60  0.05 -0.01 0.50 0.50 1.4
## A8   0.05 -0.02  0.57  0.05  0.02 0.36 0.64 1.0
## A9   0.04  0.10  0.70  0.03  0.04 0.51 0.49 1.1
## A10  0.28 -0.08  0.33  0.11  0.06 0.31 0.69 2.4
## C1   0.02 -0.02 -0.04  0.60  0.12 0.39 0.61 1.1
## C2   0.05  0.05  0.08 -0.54  0.12 0.31 0.69 1.2
## C3  -0.08  0.05  0.07  0.40  0.27 0.24 0.76 2.0
## C4  -0.02  0.30  0.01 -0.53  0.02 0.45 0.55 1.6
## C5   0.08  0.00  0.01  0.63 -0.08 0.41 0.59 1.1
## C6   0.01  0.10  0.05 -0.59  0.06 0.37 0.63 1.1
## C7  -0.06  0.14  0.00  0.56  0.05 0.30 0.70 1.2
## C8  -0.02  0.17 -0.11 -0.45 -0.03 0.30 0.70 1.4
## C9   0.05  0.11  0.04  0.64 -0.03 0.40 0.60 1.1
## C10  0.00  0.06  0.02  0.47  0.23 0.28 0.72 1.5
## O1  -0.03 -0.03 -0.04  0.02  0.60 0.36 0.64 1.0
## O2   0.07  0.22 -0.02  0.05 -0.55 0.36 0.64 1.4
## O3  -0.02  0.10  0.07 -0.10  0.53 0.30 0.70 1.2
## O4   0.09  0.14 -0.12  0.11 -0.47 0.26 0.74 1.5
## O5   0.15 -0.01 -0.06  0.14  0.58 0.42 0.58 1.3
## O6  -0.04  0.04 -0.08  0.07 -0.49 0.26 0.74 1.1
## O7   0.02 -0.10 -0.03  0.17  0.49 0.30 0.70 1.3
## O8  -0.04  0.08 -0.11 -0.06  0.57 0.33 0.67 1.2
## O9  -0.19  0.15  0.20  0.04  0.35 0.20 0.80 2.7
## O10  0.13  0.02  0.00  0.02  0.66 0.48 0.52 1.1
##
##                        PA1  PA3  PA5  PA2  PA4
## SS loadings           5.06 4.55 3.72 3.27 3.20
```

```
## Proportion Var        0.10 0.09 0.07 0.07 0.06
## Cumulative Var        0.10 0.19 0.27 0.33 0.40
## Proportion Explained  0.26 0.23 0.19 0.16 0.16
## Cumulative Proportion 0.26 0.49 0.67 0.84 1.00
##
##  With factor correlations of
##       PA1   PA3   PA5   PA2   PA4
## PA1  1.00 -0.23  0.25  0.09  0.17
## PA3 -0.23  1.00 -0.03 -0.24 -0.08
## PA5  0.25 -0.03  1.00  0.14  0.07
## PA2  0.09 -0.24  0.14  1.00  0.05
## PA4  0.17 -0.08  0.07  0.05  1.00
##
## Mean item complexity =  1.3
## Test of the hypothesis that 5 factors are sufficient.
##
## df null model =  1225  with the objective function =  19.13 with Chi Square =  376850.5
## df of  the model are 985  and the objective function was  3.02
##
## The root mean square of the residuals (RMSR) is  0.03
## The df corrected root mean square of the residuals is  0.04
##
## The harmonic n.obs is  19719 with the empirical chi square  26611.69  with prob <  0
## The total n.obs was  19719  with Likelihood Chi Square =  59449.71  with prob <  0
##
## Tucker Lewis Index of factoring reliability =  0.806
## RMSEA index =  0.055  and the 90 % confidence intervals are  0.054 0.055
## BIC =  49708.71
## Fit based upon off diagonal values = 0.97
## Measures of factor score adequacy
##                                                 PA1  PA3  PA5  PA2  PA4
## Correlation of (regression) scores with factors   0.93 0.93 0.92 0.90 0.90
## Multiple R square of scores with factors          0.86 0.86 0.84 0.80 0.82
## Minimum correlation of possible factor scores     0.72 0.72 0.69 0.61 0.63
```

```
print(fitBig5_fiveFac$loadings, cutoff = 0.3, sort = FALSE)
```

```
##
## Loadings:
##     PA1    PA3    PA5    PA2    PA4
## E1   0.691
## E2  -0.695
## E3   0.632
## E4  -0.720
## E5   0.723
## E6  -0.544
## E7   0.738
## E8  -0.595
## E9   0.639
## E10 -0.652
## N1          0.688
## N2         -0.501
## N3          0.614
## N4
```

```
## N5          0.535
## N6          0.747
## N7          0.710
## N8          0.742
## N9          0.729
## N10         0.579
## A1                -0.437
## A2                 0.502
## A3                -0.407
## A4                 0.800
## A5                -0.658
## A6                 0.607
## A7                -0.601
## A8                 0.575
## A9                 0.696
## A10                0.335
## C1                        0.600
## C2                       -0.537
## C3                        0.401
## C4                       -0.531
## C5                        0.632
## C6                       -0.586
## C7                        0.557
## C8                       -0.450
## C9                        0.642
## C10                       0.472
## O1                              0.600
## O2                             -0.554
## O3                              0.531
## O4                             -0.473
## O5                              0.582
## O6                             -0.490
## O7                              0.490
## O8                              0.565
## O9                              0.350
## O10                             0.662
##
##                  PA1   PA3   PA5   PA2   PA4
## SS loadings    4.890 4.419 3.616 3.181 3.156
## Proportion Var 0.098 0.088 0.072 0.064 0.063
## Cumulative Var 0.098 0.186 0.258 0.322 0.385
```

```
#do the factor analysis for 6
fitBig5_sixFac<-fa(Big5Data,nfactors = 6, rotate = "oblimin", fm = 'pa')
fitBig5_sixFac
```

```
## Factor Analysis using method =  pa
## Call: fa(r = Big5Data, nfactors = 6, rotate = "oblimin", fm = "pa")
## Standardized loadings (pattern matrix) based upon correlation matrix
##       PA5   PA2   PA4   PA3   PA6   PA1   h2   u2 com
## E1  -0.66  0.01  0.00 -0.02  0.02  0.18 0.48 0.52 1.2
## E2   0.73 -0.10 -0.02  0.03  0.01  0.13 0.51 0.49 1.1
## E3  -0.61 -0.19  0.18  0.08 -0.04  0.13 0.58 0.42 1.5
## E4   0.75  0.04  0.06 -0.01  0.04  0.11 0.55 0.45 1.1
```

```
## E5  -0.71  0.02  0.12  0.07  0.04  0.04 0.59 0.41 1.1
## E6   0.60 -0.01 -0.05 -0.02 -0.17  0.22 0.47 0.53 1.5
## E7  -0.72 -0.02  0.07  0.01  0.00  0.09 0.57 0.43 1.1
## E8   0.59 -0.04  0.11  0.07  0.00 -0.01 0.33 0.67 1.1
## E9  -0.61  0.03 -0.07 -0.03  0.10  0.15 0.41 0.59 1.2
## E10  0.67  0.09  0.04 -0.01  0.02  0.09 0.47 0.53 1.1
## N1   0.03  0.71  0.08  0.05 -0.05 -0.11 0.51 0.49 1.1
## N2   0.00 -0.56  0.04 -0.12  0.08  0.31 0.39 0.61 1.7
## N3   0.09  0.62  0.19  0.10  0.01 -0.08 0.43 0.57 1.3
## N4  -0.08 -0.33 -0.03  0.05 -0.05  0.26 0.20 0.80 2.1
## N5   0.00  0.52  0.02 -0.07 -0.10  0.11 0.33 0.67 1.2
## N6   0.00  0.74  0.07 -0.02 -0.06  0.05 0.57 0.43 1.0
## N7  -0.05  0.70 -0.04 -0.10  0.03  0.10 0.53 0.47 1.1
## N8  -0.04  0.73 -0.04 -0.10  0.02  0.09 0.58 0.42 1.1
## N9  -0.03  0.72 -0.15  0.03  0.01  0.05 0.53 0.47 1.1
## N10  0.19  0.59  0.01 -0.10  0.07 -0.09 0.47 0.53 1.3
## A1   0.00  0.06 -0.41  0.00 -0.04  0.30 0.27 0.73 1.9
## A2  -0.28 -0.04  0.50 -0.04  0.05 -0.02 0.41 0.59 1.6
## A3  -0.16  0.26 -0.40 -0.16  0.10  0.06 0.27 0.73 2.7
## A4   0.07  0.02  0.81 -0.01  0.00  0.04 0.63 0.37 1.0
## A5   0.10  0.02 -0.63  0.04  0.02  0.20 0.48 0.52 1.3
## A6   0.08  0.11  0.63  0.00 -0.07  0.13 0.41 0.59 1.2
## A7   0.27  0.07 -0.58  0.03  0.01  0.20 0.54 0.46 1.8
## A8  -0.04 -0.03  0.58  0.05  0.02  0.04 0.37 0.63 1.0
## A9  -0.01  0.08  0.71  0.03  0.05  0.09 0.54 0.46 1.1
## A10 -0.24 -0.11  0.37  0.10  0.09  0.21 0.36 0.64 2.9
## C1   0.00 -0.03 -0.03  0.59  0.13  0.07 0.39 0.61 1.1
## C2  -0.03  0.02  0.09 -0.55  0.12  0.09 0.32 0.68 1.2
## C3   0.09  0.05  0.08  0.39  0.28  0.06 0.25 0.75 2.1
## C4   0.04  0.28  0.02 -0.54  0.03  0.09 0.46 0.54 1.6
## C5  -0.05 -0.01  0.03  0.62 -0.07  0.13 0.42 0.58 1.1
## C6   0.02  0.07  0.07 -0.61  0.07  0.15 0.41 0.59 1.2
## C7   0.05  0.15  0.00  0.56  0.06 -0.01 0.30 0.70 1.2
## C8   0.04  0.15 -0.10 -0.46 -0.02  0.12 0.32 0.68 1.5
## C9  -0.04  0.11  0.05  0.64 -0.02  0.06 0.40 0.60 1.1
## C10  0.02  0.05  0.04  0.46  0.25  0.10 0.29 0.71 1.7
## O1   0.00 -0.01 -0.06  0.03  0.58 -0.19 0.38 0.62 1.2
## O2  -0.02  0.19  0.01  0.03 -0.54  0.28 0.42 0.58 1.8
## O3   0.04  0.08  0.09 -0.11  0.54  0.10 0.32 0.68 1.3
## O4  -0.05  0.11 -0.09  0.09 -0.45  0.27 0.31 0.69 2.0
## O5  -0.11 -0.03 -0.03  0.12  0.61  0.20 0.47 0.53 1.4
## O6   0.03  0.05 -0.08  0.07 -0.49 -0.01 0.27 0.73 1.1
## O7   0.00 -0.11 -0.02  0.16  0.50  0.05 0.31 0.69 1.3
## O8   0.01  0.11 -0.13 -0.05  0.55 -0.17 0.34 0.66 1.4
## O9   0.18  0.16  0.20  0.04  0.35 -0.05 0.19 0.81 2.8
## O10 -0.09  0.00  0.03  0.00  0.68  0.15 0.52 0.48 1.1
##
##                        PA5  PA2  PA4  PA3  PA6  PA1
## SS loadings           5.03 4.56 3.73 3.29 3.23 1.03
## Proportion Var        0.10 0.09 0.07 0.07 0.06 0.02
## Cumulative Var        0.10 0.19 0.27 0.33 0.40 0.42
## Proportion Explained  0.24 0.22 0.18 0.16 0.15 0.05
## Cumulative Proportion 0.24 0.46 0.64 0.80 0.95 1.00
##
```

```
##  With factor correlations of
##       PA5   PA2   PA4   PA3   PA6   PA1
## PA5  1.00  0.24 -0.25 -0.09 -0.17 -0.05
## PA2  0.24  1.00 -0.03 -0.24 -0.08  0.06
## PA4 -0.25 -0.03  1.00  0.13  0.07 -0.02
## PA3 -0.09 -0.24  0.13  1.00  0.04  0.01
## PA6 -0.17 -0.08  0.07  0.04  1.00 -0.01
## PA1 -0.05  0.06 -0.02  0.01 -0.01  1.00
##
## Mean item complexity =  1.4
## Test of the hypothesis that 6 factors are sufficient.
##
## df null model =  1225  with the objective function =  19.13 with Chi Square =  376850.5
## df of  the model are 940  and the objective function was  2.39
##
## The root mean square of the residuals (RMSR) is  0.03
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  19719 with the empirical chi square  17712.86  with prob <  0
## The total n.obs was  19719  with Likelihood Chi Square =  47036.98  with prob <  0
##
## Tucker Lewis Index of factoring reliability =  0.84
## RMSEA index =  0.05  and the 90 % confidence intervals are  0.049 0.05
## BIC =  37741
## Fit based upon off diagonal values = 0.98
## Measures of factor score adequacy
##                                                  PA5  PA2  PA4  PA3  PA6  PA1
## Correlation of (regression) scores with factors   0.93 0.93 0.92 0.90 0.91 0.79
## Multiple R square of scores with factors          0.86 0.86 0.85 0.81 0.83 0.63
## Minimum correlation of possible factor scores     0.72 0.72 0.70 0.62 0.65 0.25
```

```
print(fitBig5_sixFac$loadings, cutoff = 0.3, sort = FALSE)
```

```
##
## Loadings:
##     PA5    PA2    PA4    PA3    PA6    PA1
## E1  -0.658
## E2   0.728
## E3  -0.605
## E4   0.750
## E5  -0.714
## E6   0.595
## E7  -0.719
## E8   0.594
## E9  -0.611
## E10  0.672
## N1          0.707
## N2         -0.563                       0.313
## N3          0.624
## N4         -0.334
## N5          0.520
## N6          0.739
## N7          0.695
## N8          0.728
```

```
## N9          0.723
## N10         0.590
## A1                -0.405                0.303
## A2                 0.498
## A3                -0.399
## A4                 0.807
## A5                -0.634
## A6                 0.632
## A7                -0.577
## A8                 0.581
## A9                 0.714
## A10                0.368
## C1                        0.594
## C2                       -0.550
## C3                        0.394
## C4                       -0.544
## C5                        0.625
## C6                       -0.607
## C7                        0.555
## C8                       -0.465
## C9                        0.635
## C10                       0.462
## O1                                0.584
## O2                               -0.536
## O3                                0.544
## O4                               -0.453
## O5                                0.612
## O6                               -0.491
## O7                                0.497
## O8                                0.550
## O9                                0.345
## O10                               0.685
##
##                  PA5   PA2   PA4   PA3   PA6   PA1
## SS loadings    4.844 4.427 3.614 3.202 3.177 1.022
## Proportion Var 0.097 0.089 0.072 0.064 0.064 0.020
## Cumulative Var 0.097 0.185 0.258 0.322 0.385 0.406
```

```
#do the factor analysis for 7
fitBig5_sevFac<-fa(Big5Data,nfactors = 7, rotate = "oblimin", fm = 'pa')
fitBig5_sevFac
```

```
## Factor Analysis using method =  pa
## Call: fa(r = Big5Data, nfactors = 7, rotate = "oblimin", fm = "pa")
## Standardized loadings (pattern matrix) based upon correlation matrix
##       PA3   PA5   PA1   PA4   PA6   PA7   PA2   h2   u2 com
## E1  -0.66  0.01 -0.01 -0.02  0.05 -0.04  0.17 0.48 0.52 1.2
## E2   0.73 -0.10 -0.02  0.03  0.02 -0.03  0.11 0.51 0.49 1.1
## E3  -0.60 -0.19  0.18  0.07 -0.04 -0.02  0.15 0.59 0.41 1.6
## E4   0.75  0.03  0.06 -0.02  0.02  0.03  0.13 0.56 0.44 1.1
## E5  -0.71  0.01  0.13  0.07  0.00  0.05  0.07 0.60 0.40 1.1
## E6   0.59 -0.01 -0.06 -0.02 -0.09 -0.14  0.20 0.47 0.53 1.4
## E7  -0.72 -0.02  0.08  0.01 -0.02  0.02  0.11 0.58 0.42 1.1
## E8   0.59 -0.05  0.12  0.06 -0.08  0.08  0.03 0.34 0.66 1.2
```

```
## E9  -0.61  0.03 -0.08 -0.03  0.13 -0.02  0.12 0.41 0.59 1.2
## E10  0.67  0.08  0.04 -0.01  0.01  0.01  0.09 0.47 0.53 1.1
## N1   0.03  0.70  0.09  0.05 -0.09  0.05 -0.08 0.51 0.49 1.1
## N2   0.00 -0.57  0.03 -0.12  0.08 -0.03  0.31 0.39 0.61 1.7
## N3   0.09  0.62  0.20  0.10 -0.03  0.06 -0.04 0.43 0.57 1.3
## N4  -0.08 -0.34 -0.04  0.05 -0.01 -0.07  0.25 0.20 0.80 2.2
## N5   0.01  0.52  0.01 -0.07 -0.03 -0.08  0.10 0.33 0.67 1.2
## N6   0.00  0.74  0.07 -0.02 -0.02 -0.05  0.04 0.57 0.43 1.0
## N7  -0.05  0.70 -0.05 -0.09  0.09 -0.06  0.07 0.53 0.47 1.1
## N8  -0.04  0.74 -0.06 -0.09  0.09 -0.07  0.05 0.58 0.42 1.1
## N9  -0.03  0.72 -0.14  0.02 -0.03  0.05  0.07 0.53 0.47 1.1
## N10  0.19  0.59  0.01 -0.10  0.04  0.06 -0.08 0.47 0.53 1.3
## A1   0.01  0.06 -0.42  0.00  0.07 -0.15  0.25 0.27 0.73 2.0
## A2  -0.28 -0.05  0.51 -0.05 -0.01  0.08  0.02 0.42 0.58 1.7
## A3  -0.16  0.24 -0.39 -0.17 -0.02  0.15  0.11 0.29 0.71 3.1
## A4   0.07  0.02  0.80 -0.01  0.02 -0.03  0.04 0.63 0.37 1.0
## A5   0.10  0.02 -0.64  0.04  0.04 -0.02  0.18 0.48 0.52 1.2
## A6   0.08  0.11  0.62  0.01  0.04 -0.14  0.11 0.41 0.59 1.3
## A7   0.27  0.07 -0.59  0.03  0.03 -0.03  0.19 0.54 0.46 1.7
## A8  -0.04 -0.04  0.59  0.04 -0.02  0.04  0.07 0.38 0.62 1.1
## A9  -0.01  0.09  0.70  0.03  0.10 -0.05  0.08 0.53 0.47 1.1
## A10 -0.24 -0.11  0.36  0.10  0.11 -0.04  0.20 0.36 0.64 3.1
## C1  -0.01 -0.04 -0.01  0.58  0.00  0.15  0.13 0.40 0.60 1.2
## C2  -0.03  0.00  0.11 -0.57 -0.03  0.19  0.17 0.37 0.63 1.5
## C3   0.09  0.03  0.09  0.39  0.14  0.18  0.11 0.25 0.75 2.1
## C4   0.04  0.26  0.03 -0.56 -0.04  0.09  0.13 0.48 0.52 1.6
## C5  -0.05  0.00  0.02  0.62  0.00 -0.11  0.11 0.42 0.58 1.1
## C6   0.02  0.05  0.08 -0.63 -0.01  0.11  0.20 0.44 0.56 1.3
## C7   0.05  0.13  0.02  0.55 -0.06  0.12  0.05 0.30 0.70 1.3
## C8   0.04  0.14 -0.11 -0.47  0.00 -0.03  0.11 0.32 0.68 1.5
## C9  -0.04  0.10  0.07  0.62 -0.08  0.04  0.11 0.41 0.59 1.2
## C10  0.02  0.03  0.06  0.46  0.09  0.19  0.16 0.30 0.70 1.8
## O1  -0.01 -0.06 -0.02  0.01  0.06  0.71 -0.02 0.55 0.45 1.0
## O2  -0.02  0.18  0.01  0.00 -0.37 -0.27  0.28 0.42 0.58 3.3
## O3   0.05  0.11  0.06 -0.08  0.63  0.00  0.01 0.40 0.60 1.1
## O4  -0.05  0.10 -0.09  0.06 -0.38 -0.17  0.30 0.33 0.67 2.8
## O5  -0.11 -0.04 -0.04  0.14  0.52  0.17  0.18 0.47 0.53 1.8
## O6   0.02  0.01 -0.04  0.02 -0.68  0.11  0.13 0.44 0.56 1.1
## O7  -0.01 -0.13  0.00  0.16  0.25  0.34  0.12 0.32 0.68 3.0
## O8   0.00  0.06 -0.09 -0.07  0.03  0.70  0.00 0.53 0.47 1.1
## O9   0.18  0.14  0.21  0.04  0.18  0.23 -0.01 0.19 0.81 4.7
## O10 -0.09  0.00  0.00  0.04  0.65  0.13  0.09 0.56 0.44 1.2
##
##                        PA3  PA5  PA1  PA4  PA6  PA7  PA2
## SS loadings           5.02 4.57 3.74 3.31 2.31 1.83 0.99
## Proportion Var        0.10 0.09 0.07 0.07 0.05 0.04 0.02
## Cumulative Var        0.10 0.19 0.27 0.33 0.38 0.42 0.44
## Proportion Explained  0.23 0.21 0.17 0.15 0.11 0.08 0.05
## Cumulative Proportion 0.23 0.44 0.61 0.76 0.87 0.95 1.00
##
##  With factor correlations of
##       PA3   PA5   PA1   PA4   PA6   PA7   PA2
## PA3  1.00  0.24 -0.25 -0.08 -0.17 -0.07 -0.05
## PA5  0.24  1.00 -0.03 -0.24 -0.09 -0.02  0.07
```

```
## PA1 -0.25 -0.03  1.00  0.13  0.09  0.01 -0.01
## PA4 -0.08 -0.24  0.13  1.00  0.02  0.03  0.01
## PA6 -0.17 -0.09  0.09  0.02  1.00  0.40  0.04
## PA7 -0.07 -0.02  0.01  0.03  0.40  1.00 -0.04
## PA2 -0.05  0.07 -0.01  0.01  0.04 -0.04  1.00
##
## Mean item complexity =  1.6
## Test of the hypothesis that 7 factors are sufficient.
##
## df null model =  1225  with the objective function =  19.13 with Chi Square =  376850.5
## df of  the model are 896  and the objective function was  1.95
##
## The root mean square of the residuals (RMSR) is  0.02
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  19719 with the empirical chi square  12560.06  with prob <  0
## The total n.obs was  19719  with Likelihood Chi Square =  38453.57  with prob <  0
##
## Tucker Lewis Index of factoring reliability =  0.863
## RMSEA index =  0.046  and the 90 % confidence intervals are  0.046 0.047
## BIC =  29592.72
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                  PA3  PA5  PA1  PA4  PA6  PA7
## Correlation of (regression) scores with factors   0.93 0.93 0.92 0.90 0.83 0.82
## Multiple R square of scores with factors          0.86 0.86 0.85 0.81 0.69 0.67
## Minimum correlation of possible factor scores     0.72 0.71 0.70 0.62 0.38 0.35
##                                                  PA2
## Correlation of (regression) scores with factors   0.78
## Multiple R square of scores with factors          0.61
## Minimum correlation of possible factor scores     0.22
```

```
print(fitBig5_sevFac$loadings, cutoff = 0.3, sort = TRUE)
```

```
##
## Loadings:
##     PA3    PA5    PA1    PA4    PA6    PA7    PA2
## E1  -0.656
## E2   0.726
## E3  -0.605
## E4   0.749
## E5  -0.713
## E6   0.594
## E7  -0.719
## E8   0.593
## E9  -0.609
## E10  0.671
## N1          0.704
## N2         -0.573                            0.308
## N3          0.620
## N5          0.522
## N6          0.740
## N7          0.703
## N8          0.737
```

```
## N9          0.716
## N10         0.592
## A2                 0.507
## A4                 0.804
## A5                -0.641
## A6                 0.618
## A7                -0.585
## A8                 0.588
## A9                 0.704
## C1                        0.585
## C2                       -0.574
## C4                       -0.557
## C5                        0.624
## C6                       -0.626
## C7                        0.546
## C9                        0.624
## O3                               0.627
## O5                               0.517
## O6                              -0.685
## O10                              0.655
## O1                                      0.709
## O8                                      0.705
## N4          -0.340
## A1                -0.425
## A3                -0.389
## A10                0.359
## C3                        0.393
## C8                       -0.465
## C10                       0.457
## O2                              -0.373
## O4                              -0.376
## O7                                      0.338
## O9
##
##                  PA3   PA5   PA1   PA4   PA6   PA7   PA2
## SS loadings    4.827 4.428 3.617 3.213 2.082 1.649 0.982
## Proportion Var 0.097 0.089 0.072 0.064 0.042 0.033 0.020
## Cumulative Var 0.097 0.185 0.257 0.322 0.363 0.396 0.416
```

```
# #do the factor analysis for 4
fitBig5_fourFac<-fa(Big5Data,nfactors = 4, rotate = "oblimin", fm = 'pa')
fitBig5_fourFac
```

```
## Factor Analysis using method =  pa
## Call: fa(r = Big5Data, nfactors = 4, rotate = "oblimin", fm = "pa")
## Standardized loadings (pattern matrix) based upon correlation matrix
##       PA4   PA1   PA3   PA2   h2   u2 com
## E1  -0.64  0.00  0.00  0.03 0.41 0.59 1.0
## E2   0.66 -0.07 -0.05 -0.02 0.44 0.56 1.0
## E3  -0.62 -0.23  0.19 -0.04 0.57 0.43 1.5
## E4   0.67  0.10  0.02  0.01 0.48 0.52 1.0
## E5  -0.67 -0.05  0.17  0.06 0.55 0.45 1.1
## E6   0.54  0.02 -0.08 -0.21 0.40 0.60 1.3
## E7  -0.70 -0.05  0.09  0.01 0.54 0.46 1.0
```

```
## E8   0.55 -0.03  0.10 -0.02 0.29 0.71 1.1
## E9  -0.58  0.01 -0.06  0.11 0.35 0.65 1.1
## E10  0.62  0.13  0.02 -0.01 0.43 0.57 1.1
## N1   0.15  0.60  0.22 -0.04 0.44 0.56 1.4
## N2  -0.17 -0.40 -0.13  0.04 0.22 0.78 1.6
## N3   0.19  0.52  0.32  0.02 0.40 0.60 1.9
## N4  -0.13 -0.33 -0.06 -0.06 0.14 0.86 1.4
## N5   0.05  0.51  0.06 -0.11 0.29 0.71 1.1
## N6   0.09  0.68  0.17 -0.06 0.51 0.49 1.2
## N7   0.03  0.68  0.03  0.02 0.46 0.54 1.0
## N8   0.04  0.70  0.02  0.01 0.50 0.50 1.0
## N9   0.12  0.60  0.00  0.01 0.40 0.60 1.1
## N10  0.24  0.61  0.04  0.07 0.47 0.53 1.3
## A1   0.07  0.02 -0.34 -0.05 0.14 0.86 1.1
## A2  -0.40  0.03  0.41  0.05 0.39 0.61 2.0
## A3  -0.08  0.28 -0.39  0.10 0.23 0.77 2.1
## A4  -0.13  0.11  0.67 -0.03 0.49 0.51 1.1
## A5   0.23 -0.06 -0.52  0.02 0.36 0.64 1.4
## A6  -0.06  0.17  0.55 -0.10 0.33 0.67 1.3
## A7   0.38  0.01 -0.48  0.00 0.44 0.56 1.9
## A8  -0.16  0.01  0.52  0.01 0.32 0.68 1.2
## A9  -0.16  0.14  0.63  0.03 0.46 0.54 1.3
## A10 -0.32 -0.12  0.35  0.07 0.31 0.69 2.3
## C1   0.19 -0.38  0.28  0.17 0.26 0.74 2.9
## C2  -0.24  0.38 -0.22  0.07 0.20 0.80 2.5
## C3   0.20 -0.17  0.27  0.30 0.21 0.79 3.4
## C4  -0.12  0.59 -0.23 -0.02 0.40 0.60 1.4
## C5   0.14 -0.38  0.34 -0.02 0.26 0.74 2.3
## C6  -0.21  0.45 -0.26  0.01 0.27 0.73 2.1
## C7   0.25 -0.21  0.31  0.10 0.18 0.82 2.9
## C8  -0.09  0.41 -0.32 -0.06 0.28 0.72 2.0
## C9   0.17 -0.29  0.39  0.03 0.24 0.76 2.3
## C10  0.16 -0.22  0.28  0.27 0.21 0.79 3.5
## O1   0.03  0.00 -0.05  0.60 0.36 0.64 1.0
## O2   0.00  0.13  0.06 -0.54 0.32 0.68 1.1
## O3  -0.03  0.19  0.00  0.52 0.29 0.71 1.3
## O4   0.01  0.01 -0.01 -0.45 0.21 0.79 1.0
## O5  -0.09 -0.07  0.01  0.60 0.41 0.59 1.1
## O6   0.09 -0.04 -0.01 -0.48 0.25 0.75 1.1
## O7   0.03 -0.17  0.03  0.51 0.30 0.70 1.2
## O8   0.05  0.14 -0.13  0.57 0.33 0.67 1.2
## O9   0.16  0.17  0.20  0.35 0.19 0.81 2.6
## O10 -0.12  0.04 -0.01  0.67 0.48 0.52 1.1
##
##                        PA4  PA1  PA3  PA2
## SS loadings           5.24 5.13 3.81 3.27
## Proportion Var        0.10 0.10 0.08 0.07
## Cumulative Var        0.10 0.21 0.28 0.35
## Proportion Explained  0.30 0.29 0.22 0.19
## Cumulative Proportion 0.30 0.59 0.81 1.00
##
##  With factor correlations of
##       PA4   PA1   PA3   PA2
## PA4  1.00  0.19 -0.17 -0.16
```

```
## PA1  0.19  1.00 -0.06 -0.11
## PA3 -0.17 -0.06  1.00  0.08
## PA2 -0.16 -0.11  0.08  1.00
##
## Mean item complexity =  1.6
## Test of the hypothesis that 4 factors are sufficient.
##
## df null model =  1225  with the objective function =  19.13 with Chi Square =  376850.5
## df of  the model are 1031  and the objective function was  5.02
##
## The root mean square of the residuals (RMSR) is  0.05
## The df corrected root mean square of the residuals is  0.06
##
## The harmonic n.obs is  19719 with the empirical chi square  71643.15  with prob <  0
## The total n.obs was  19719  with Likelihood Chi Square =  98827.22  with prob <  0
##
## Tucker Lewis Index of factoring reliability =  0.691
## RMSEA index =  0.069  and the 90 % confidence intervals are  0.069 0.07
## BIC =  88631.31
## Fit based upon off diagonal values = 0.92
## Measures of factor score adequacy
##                                                  PA4  PA1  PA3  PA2
## Correlation of (regression) scores with factors   0.93 0.94 0.91 0.90
## Multiple R square of scores with factors          0.86 0.88 0.84 0.82
## Minimum correlation of possible factor scores     0.73 0.76 0.67 0.63
```

```
print(fitBig5_fourFac$loadings, cutoff = 0.3, sort = FALSE)
```

```
##
## Loadings:
##     PA4    PA1    PA3    PA2
## E1  -0.635
## E2   0.663
## E3  -0.625
## E4   0.671
## E5  -0.674
## E6   0.537
## E7  -0.698
## E8   0.551
## E9  -0.577
## E10  0.621
## N1          0.598
## N2         -0.399
## N3          0.521  0.317
## N4         -0.330
## N5          0.508
## N6          0.675
## N7          0.676
## N8          0.704
## N9          0.603
## N10         0.609
## A1                -0.344
## A2  -0.398         0.408
## A3                -0.386
```

```
## A4                0.670
## A5               -0.523
## A6                0.550
## A7   0.385       -0.479
## A8                0.518
## A9                0.629
## A10 -0.317        0.347
## C1         -0.377
## C2          0.376
## C3
## C4          0.594
## C5         -0.383  0.344
## C6          0.448
## C7                 0.313
## C8          0.414 -0.318
## C9                 0.390
## C10
## O1                        0.604
## O2                       -0.542
## O3                        0.522
## O4                       -0.450
## O5                        0.601
## O6                       -0.484
## O7                        0.507
## O8                        0.565
## O9                        0.345
## O10                       0.670
##
##                  PA4   PA1   PA3   PA2
## SS loadings    5.066 5.038 3.702 3.206
## Proportion Var 0.101 0.101 0.074 0.064
## Cumulative Var 0.101 0.202 0.276 0.340
```